

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I Eshwanth C(192321062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Eshwanth C

Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.
 - a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
 - b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.
2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.
 - a. Deduce the most likely original message.
 - b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.
3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.
 - a. Demonstrate how the encryption process transforms the message step-by-step.
 - b. Explain how the repeating keyword influences each character of the ciphertext.
4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.
 - a. Analyze how investigators could detect the presence of hidden data within the image.
 - b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.
5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.
 - a. Determine the most probable original message by reversing the encryption process.
 - b. Critically evaluate why such a method is vulnerable in modern communication systems.
 - c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Question 1:

Scenario: Event Communication Security

A student organization wants to securely share confidential event details with its core members using a **classical encryption technique**. They use the **Caesar Cipher**, where each letter is shifted by a fixed number of positions in the alphabet.

Reconstruct the encrypted message using a shift of 4 positions

Plaintext (Original Message):

MEET AT NOON

Using a **Caesar Cipher with a shift of +4**, each letter is replaced by the letter four positions ahead in the alphabet.

Plain Letter	Shift +4	Cipher Letter
M	Q	Q
E	I	I
E	I	I
T	X	X
A	E	E
T	X	X
N	R	R
O	S	S
O	S	S
N	R	R

Explain how an authorized member retrieves the original message

An authorized member knows that the message was encrypted using a **Caesar Cipher with a shift of 4**. To recover the original message, they perform the **reverse operation**, shifting each letter **4 positions backward** in the alphabet.

Cipher Letter	Shift -4	Original Letter
Q	M	M
I	E	E
I	E	E
X	T	T
E	A	A
X	T	T
R	N	N
S	O	O

Cipher Letter	Shift -4	Original Letter
S	O	O
R	N	N

Working Principle

1. Choose a fixed shift value (here, 4).
2. **Encryption:** Move each alphabet letter **4 positions forward**.
3. **Decryption:** Move each alphabet letter **4 positions backward** using the same key.
4. Only members who know the shift value (key) can correctly decode the message.

Question 2:

Scenario: Intercepted Suspicious Message

A cybersecurity intern intercepts the coded message:

GSRH RH Z HVXIVG

The message resembles a simple substitution cipher.

a. Deduce the most likely original message

The ciphertext "**GSRH RH Z HVXIVG**" is encrypted using the **Atbash Cipher**, where each letter is replaced by its opposite in the alphabet ($A \leftrightarrow Z$, $B \leftrightarrow Y$, $C \leftrightarrow X$, etc.).

Cipher Letter	Plain Letter
G	T
S	H
R	I
H	S
R	I
H	S
Z	A
H	S
V	E
X	C
I	R
V	E
G	T

Original Message
THIS IS A SECRET

Justify your answer

The message is identified as an **Atbash Cipher** because:

- The pattern of repeated letters is preserved.
- The word **"RH"** appears twice, which decrypts to **"IS"**, a common English word.
- The final word **"HVXIVG"** becomes **"SECRET"**, matching meaningful English text.
- Atbash uses a fixed substitution:
 - A ↔ Z
 - B ↔ Y
 - C ↔ X
 - D ↔ W
 - E ↔ V
 - ...
- Since every letter has a unique opposite, the same substitution is applied consistently throughout the message.

Conclusion: The original message is **"THIS IS A SECRET."**

Question 3:

Scenario: Secure Messaging System

A company encrypts the word **"HELLO"** using the keyword **"KEY"** (Vigenère Cipher).

a. Demonstrate the encryption process

Step 1: Write the plaintext
HELLO

Step 2: Repeat the keyword
KEYKE

Letter

A=0

H=7

E=4

L=11

L=11

O=14

Value

...

Plain

H(7)

E(4)

L(11)

L(11)

O(14)

Key

K(10)

E(4)

Y(24)

K(10)

E(4)

Ca

17

8

35

21

18

Explain the influence of the repeating keyword

- The keyword **KEY** is repeated until it matches the message length.

HELLO
KEYKE

- Each plaintext letter is shifted according to the corresponding keyword letter.
- Different keyword letters create different shifts, making the cipher stronger than the Caesar Cipher.
- Repeating the keyword means the shift pattern repeats after every three characters.

Question 4:

Scenario: Suspicious Digital Asset Investigation

Investigators suspect an image contains hidden confidential information.

a. Analyze how investigators could detect hidden data

Investigators can detect hidden information by:

1. **Steganalysis**
 - Examining image pixels for unusual patterns.
 - Detecting modifications in the Least Significant Bits (LSB).
2. **Metadata Analysis**
 - Checking image metadata for hidden comments or unexpected information.
3. **Statistical Analysis**
 - Comparing pixel distributions with normal images.
 - Looking for abnormal noise or compression artifacts.
4. **Steganography Detection Tools**
 - Using forensic software to identify embedded files or messages.

b. Recommend two techniques to improve security

1. Encrypt the secret data before hiding it

- Use AES or another strong encryption algorithm before embedding the message.

- Even if extracted, the data remains unreadable without the encryption key.

2. Randomized or Adaptive Embedding

- Hide information in randomly selected pixels instead of sequential locations.
- This makes detection using statistical analysis much more difficult.

Question 5:

Scenario: Legacy Encryption in Corporate Communication

Ciphertext:

WECRLTEERDSOEEFEAOCAIVDEN

This is encrypted using a **Rail Fence Cipher (3 rails)**.

a. Determine the original message

After reversing the 3-rail Rail Fence Cipher, the plaintext becomes:

WE ARE DISCOVERED FLEE AT ONCE

b. Why is this method vulnerable?

The Rail Fence Cipher is insecure because:

- It only rearranges letters without changing them.
- It has a very small key space.
- Attackers can easily try different rail numbers.
- Frequency analysis and pattern recognition can reveal the plaintext.
- Modern computers can crack it within seconds.

Therefore, it is **not suitable for protecting confidential corporate communication**.

c. Suggest one practical enhancement

A practical improvement is to use the **Advanced Encryption Standard (AES-256)**.

Advantages:

- Provides strong symmetric encryption.
- Resistant to brute-force attacks.
- Widely used in banking, government, and secure communications.
- Protects data confidentiality much better than transposition ciphers.

